

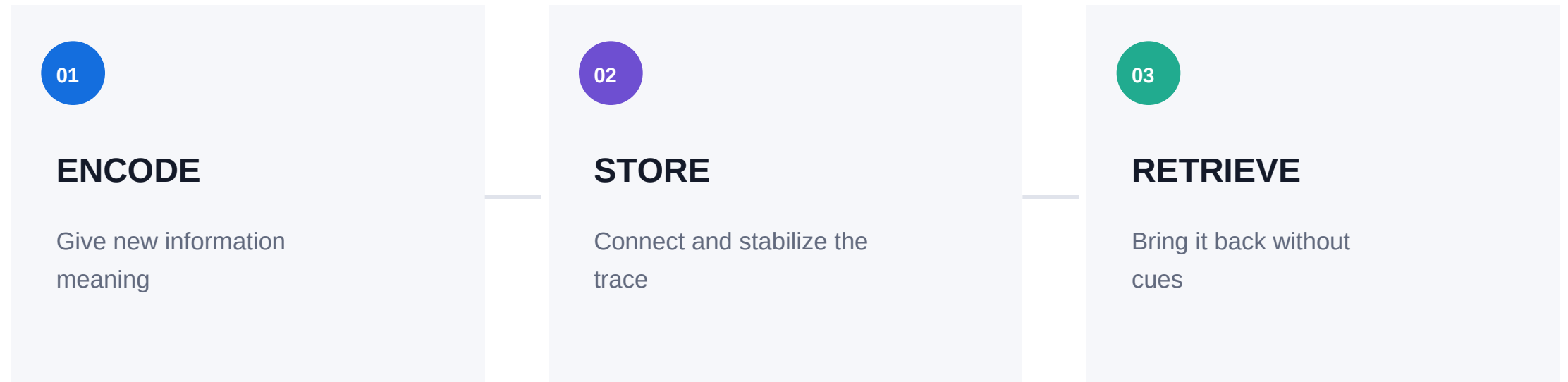
Memory & Learning Strategies

How attention, encoding, retrieval, and spacing turn information into durable knowledge.



Memory is a process

Each stage can strengthen—or lose—the information we study.



A small workspace

Working memory is powerful, but sharply limited.

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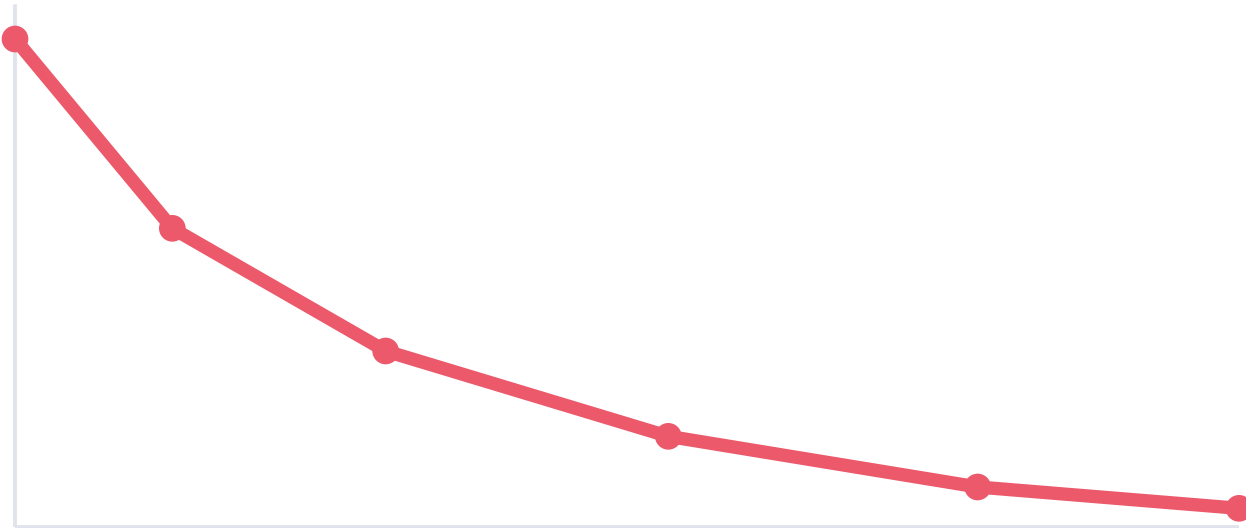
meaningful chunks

A practical estimate—not a fixed rule.

- **CHUNK**
Group details into one meaningful pattern.
- **CONNECT**
Attach the idea to something already known.
- **OFFLOAD**
Use notes and diagrams for temporary details.

Forgetting is information

It shows where access is weak—and where practice should go.

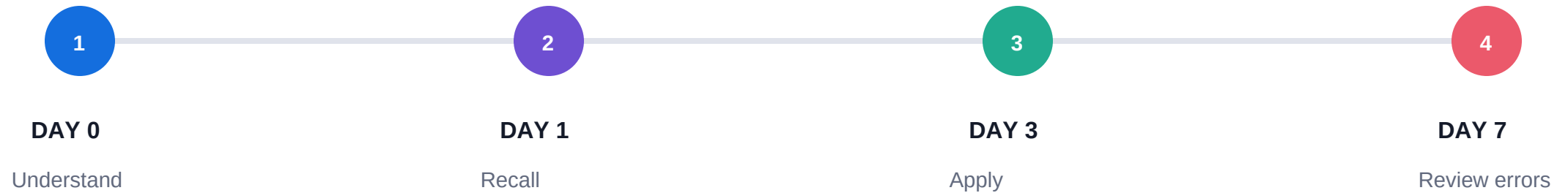


Three common causes

- Weak initial encoding
- Competing memories
- Missing retrieval cues

Space it out

Short reviews across time beat one long session.



Leave enough time to forget a little, then make retrieval effortful.

Do not just reread

Trying to recall changes memory more than another exposure.

REREADING

Feels fluent

Recognition can create an illusion of mastery.

RETRIEVAL

Builds access

Answering without notes reveals and repairs gaps.

A repeatable study loop

Use the same four moves after every lecture.

AFTER CLASS

10 min

Organize the main idea

NEXT DAY

3 prompts

Recall without notes

THREE DAYS

1 quiz

Apply the concepts

ONE WEEK

Errors

Review only weak spots

Mix related problem types

The added difficulty trains you to choose the right strategy.



Blocked practice asks: Can I repeat this method?
Interleaving asks:
Which method fits now?

Measure, then adjust

Confidence is useful only when it is checked against performance.

01 PREDICT

How well will I explain it?

02 TEST

Can I answer without cues?

03 COMPARE

Where did confidence miss?

04 ADAPT

What should the next review target?

Before the exam

A final check for durable, transferable learning.



I can explain the core idea without looking at the slide.



I can tell similar concepts apart with an example.



I have solved mixed questions, not only repeated one type.



I reviewed errors and can explain why each answer was wrong.

READY?

Recall
Apply
Explain